

```
\documentclass{article}
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```
\usepackage{amsmath}
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```
\usepackage{amssymb}
```

```
\begin{document}
```

Fréchet bounds (or triangle inequality for correlations) is given by:

```
\[
```

```
\rho_{YZ} \in
```

```
\left[
```

```
\rho_{XY} \cdot \rho_{XZ} -
```

```
\sqrt{(1 - \rho_{XY}^2)(1 - \rho_{XZ}^2)} \, ;,
```

```
\;
```

```
\rho_{XY} \cdot \rho_{XZ} +
```

```
\sqrt{(1 - \rho_{XY}^2)(1 - \rho_{XZ}^2)}
```

```
\right]
```

```
\]
```

```
\noindent
```

where:

```
\begin{itemize}
```

```
\item $\rho_{XY}$ is the degree of association between variables $X$ and $Y$,
```

```
\item $\rho_{XZ}$ is the degree of association between variables $X$ and $Z$,
```

```
\item $\rho_{YZ}$ is the degree of association between variables $Y$ and $Z$.
```

```
\end{itemize}
```

\end{document}